

Four-Track Cavitation OpenFOAM Results

Eccentric conical hydrodynamic bearing

Professor discussion results brief · 30 July 2026

Outcome

All four mechanisms were implemented and exercised with published surrogate inputs. Only the Reynolds–JFO branch passed its numerical gates. The three 3-D branches produced useful onset or failure evidence, but none produced a defensible steady quantitative solution. They are preserved and labelled; no failed branch was pushed to a cosmetic high-speed result.

Four mechanisms are separate models, not four settings of one model

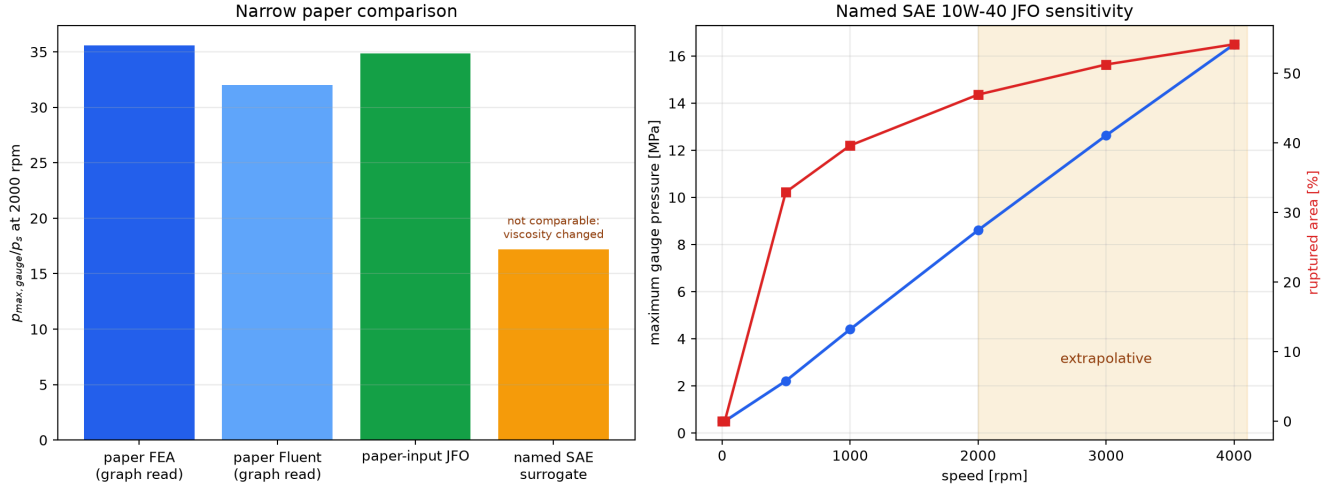
Track	Mechanism	Published surrogate	Highest state	Gate result	Status
A	Reynolds–JFO	SAE 10W-40	4000 rpm	converged + cross-code parity	ACCEPTED NUMERICAL
B	oil vapour	SAE 10W-40	28 rpm hold	phase field still drifting	UNSETTLED
C	non-condensable gas	ISO VG32	0 rpm, step 40	2.04% mass imbalance	UNSETTLED
D	atmospheric ventilation	VG22	3500 rpm, 4.37 us	p _{abs} fell below zero	REJECTED

Track	Model	Highest state	Key number	Decision
A	Reynolds–JFO	4000 rpm	$p_{g,max} = 16.495 \text{ MPa}$	Accepted numerical sensitivity
B	oil vapour	28 rpm	$\alpha_{oil,min} = 0.1112$	Unsettled phase field
C	non-condensable gas	0 rpm	2.035% imbalance	Zero-speed gate failed
D	atmospheric ventilation	3500 rpm startup	$p_{abs,min} = -31.76 \text{ MPa}$	Physically reject

Track A – Reynolds–JFO

Accepted numerical candidate, not experimental validation

The mass-conserving pressure–fill solver converged at every retained speed. Native OpenFOAM and independent Python peak pressures agree within 0.0257%. This establishes implementation parity, not correctness of the shared physics, feed model, or cavity front.



At 2000 rpm, the earlier run using the conical paper’s viscosity gives $\frac{p_{\text{max, gauge}}}{p_s} = 34.8411$. Graph reading from Gangrade et al. gives approximately 35.6 for FEA and 32.0 for Fluent: a narrow one-scalar difference of -2.13% and $+8.88\%$, respectively. This does not validate load, torque, leakage, fill fraction, or cavity geometry.

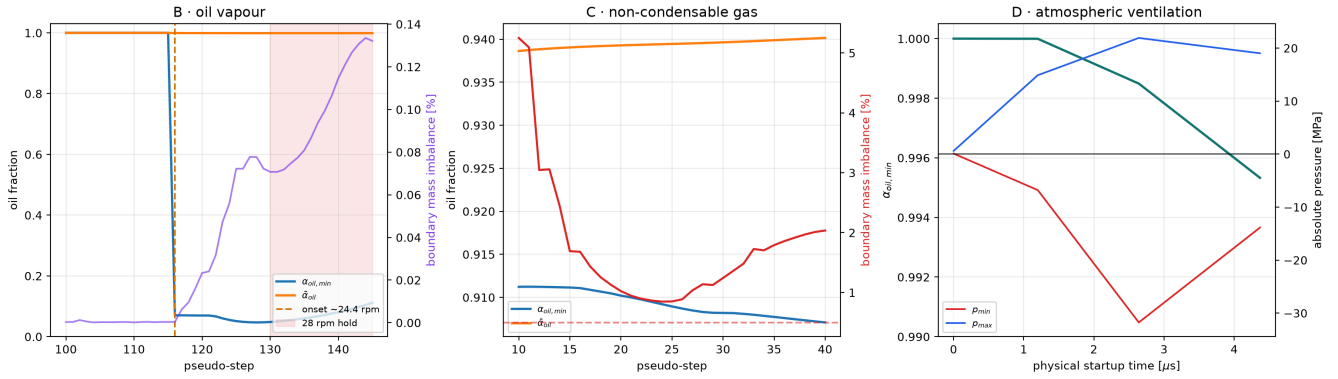
The named SAE 10W-40 surrogate gives 17.215 at 2000 rpm because its viscosity is 0.0125 Pa s rather than the direct paper’s 0.0277 Pa s. It is therefore not presented as a match. At 3000 and 4000 rpm the named-oil study gives:

Speed	$p_{g, \text{max}}$	Ruptured area	α_{min}
3000 rpm	12.6268 MPa	51.257%	0.26394
4000 rpm	16.4950 MPa	54.163%	0.26315

These two speeds are explicitly extrapolative sensitivity cases beyond the direct paper’s quoted 2000 rpm point.

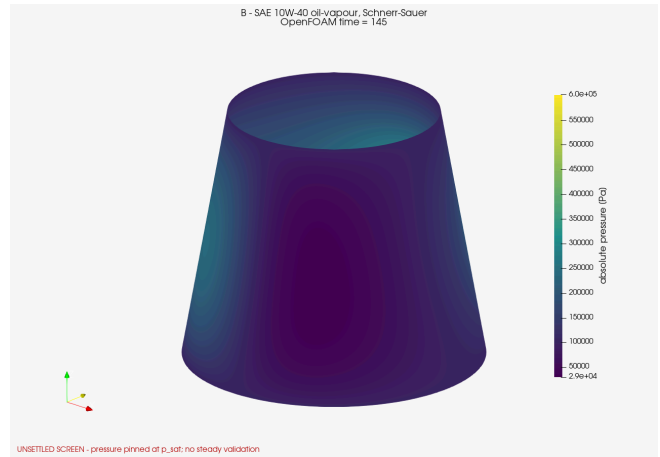
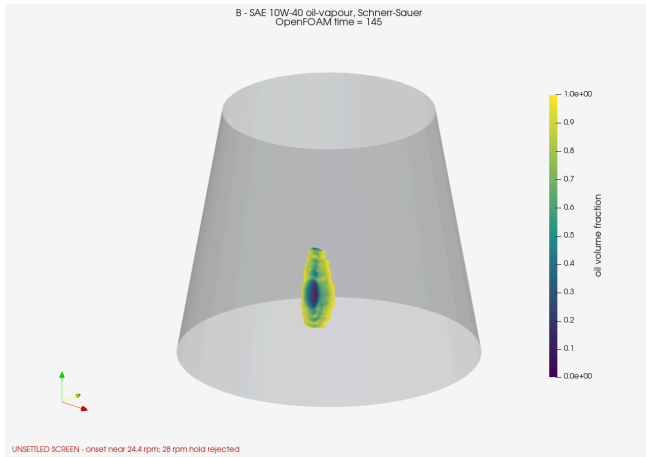
Tracks B–D – 3-D screening evidence

3-D screening diagnostics — none passed a steady quantitative gate



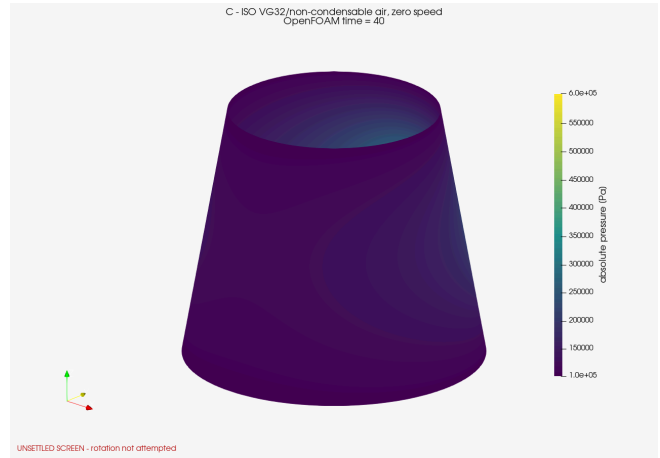
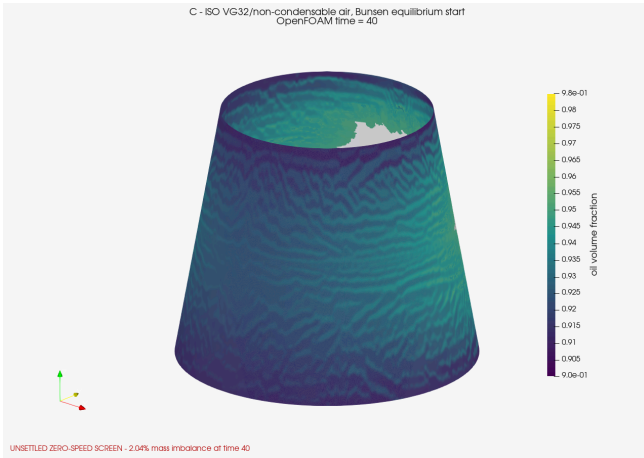
B · Oil-vapour phase change

- Native OpenFOAM Schnerr–Sauer, homogeneous VOF, SAE 10W-40 published simulation surrogate, and $p_{\text{sat}} = 29185 \text{ Pa}$.
- Stable liquid states were retained at 0 and 20 rpm.
- Phase onset occurred near 24.4 rpm. At step 145 of the 28 rpm hold, $\alpha_{\text{oil,min}} = 0.11117$, mean oil fraction was 0.998999, and boundary mass imbalance was 0.1321%.
- The minimum fraction was recovering while the mean fraction continued to fall. The phase field had not plateaued; higher speed was blocked.
- The source reports Zwart coefficients. This OpenFOAM installation has no native Zwart model, so the documented Schnerr–Sauer translation is a screen, not a reproduction.



C · Non-condensable gas / pseudo-cavitation

- Compressible VOF with ISO VG32 oil and ideal-gas air; no phase-change mass transfer.
- Oil, air, temperature, and Bunsen coefficient came from Wettmarshausen et al. The initial gas fraction was evaluated from their pressure-dependent equation, not guessed.
- At zero speed and step 40, $\alpha_{oil,min} = 0.90709$, mean oil fraction was 0.94015, and boundary mass imbalance was 2.035%.
- The declared mass gate was below 0.5%; the field was still drifting. Rotation was therefore not attempted.



D · Atmospheric ventilation

- Native incompressible VOF; pure-oil pressure feed; pure-air backflow at both axial openings; no phase-change source.
- VG22 oil/air data and measured 0.04 N/m surface tension came from Sakai, Ochiai, and Hashimoto.
- At 20 rpm over 10 microseconds, interior $\alpha_{oil,min}$ was still 0.999939: the air boundary activated, but no resolved interior tongue formed.
- At 3500 rpm, the worst startup pressures were -31.76 and $+21.91$ MPa absolute. The run was stopped because negative absolute pressure is inadmissible; vapour or released-gas physics would intervene first.
- The film-only mesh has no external reservoir, and no measured wall contact angle is available. This is not a resolved meniscus model.

Discussion decisions

Recommended statement

Reynolds–JFO is the primary model justified by the present data for mass-conserving film rupture. Oil vapour, released gas, and atmospheric ventilation remain separate hypotheses. Each needs its own material data, boundary evidence, and validation experiment before it can replace JFO or be used for quantitative conical-bearing predictions.

Questions requiring the professor’s decision

1. Is the intended observable pressure/load, a visible gas cavity, oil vapour, dissolved-air release, or ambient-air ingestion?
2. What exact V-32 product and operating temperature should be used?
3. Is the 0.5 MPa supply pressure gauge or absolute?
4. Are both axial ends flooded, sealed, or air-accessible?
5. Can the rig drawing, reservoir level, seal arrangement, and pressure-tap data be obtained?
6. For ventilation, can oil-metal contact angle be measured?

Validation route

1. Reproduce Gangrade et al.’s full normalized-pressure curve and five experimental pressure taps, retaining 496.6 and 2000 rpm as separate points.
2. Perform at least a three-grid study for peak pressure, load, feed flow, rupture boundary, and liquid deficit.
3. Validate any selected 3-D cavitation mechanism on its own fully specified experimental benchmark before applying it to the conical mesh.
4. Reserve “validated conical cavitation” for matching conical cavity data.

Sources and generated artifacts

- Gangrade et al.: [conical bearing comparison](#).
- Muchammad et al.: [SAE 10W-40 surrogate](#).
- Wettmarshausen et al.: [non-condensable-gas model](#).
- Sakai et al.: [visualized VG22 oil-air VOF](#).
- Giacomini et al.: [JFO complementarity benchmark](#).
- Miraskari et al.: [JFO numerical benchmark](#).

The complete machine-readable ledger, full-resolution figures, and animations are in `docs/handoff/media/cavitation_four_track/`. Track A media are in `docs/handoff/media/jfo_sae10w40/`. Case-specific source and result ledgers are stored beside each OpenFOAM case.